FOUNDATIONAL KNOWLEDGE IN GEOAI

FD1001: GEOAI INNOVATION CANVAS

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GEOSPATIAL- AI INNOVATION CANVAS

A Stepped-Based Methodology to the innovative & collaborative narrative

Project Name:	Stakeholder:
Designed For :	Designed by:

Cognitive Learning Theories : Cognitive Structural Theory, Cognitive Complexity Theory, Constructivist learning Approach; Innovation Theory : Computational Thinking , Systems Design Thinking, Product Design etc.

INNOVATION, KNOWLEDGE & SKILLS

PAINS

What are the real frustrations/problems they are facing?

EDUCATION & KNOWLEDGE SHARING THROUGH KEY NOTES , JOINT WORKSHOPS (USE CASES) ,BOOT CAMPS, HACKATHONS & INCUBATORS

SUSTAINABLE COMMITMENT

What sacrifices they are making & commitment to sustain & complete this project/program?

BOOT CAMPS & HACKATHONS;TECHNICAL MENTORSHIPS MENTORSHIP,JOINT WORKSHOPS, GEO AI COMMUNITIES OF PRACTICE (COP)

RETURN

What do they walk away after the project ends?
What would they expect in return after attending the project/ program?

CERTIFIED & EMPOWERED

GAINS

What do they want, need or dream ?
What benefits are they /partners seeking at the end of the project/program?

NEW KNOWLEDGE & SKILLS
RECOGNISED GEOSPATIAL-AI CERTIFICATION

ENTERING BEHAVIOUR

Before this project/program:
How would you describe behaviour of participants?
What are their knowledge, skills & attitudes?

BOOT CAMPS & HACKATHONS; MENTORSHIP, JOINT WORKSHOPs, INCUBATORS, GEO AI Communities of Practice (COP)

EXPERIENCE JOURNEY

- What chronological touch points do they experience before, during and after the project/program?
- How does this shape the intended behaviour change?

JOBS TO BE DONE

What are their' to solve/do list?

PROMISE

- How does the project/program create value for them?
- Does this promise alleviate pains and create gains?

EXITING BEHAVIOUR

As a result of this project/program, what new skills, knowledge level, attitudes and connections/networking have they acquired?

EDUCATION & KNOWLEDGE SHARING :
KEY NOTES, BOOT CAMPS & HACKATHONS;
MENTORSHIP, JOINT WORKSHOP, INCUBATORS, GEO AI Communities of Practice (COP)

INSTRUCTIONAL DESIGN

- What do they need to learn? How can they best learn?
- What skills & knowledge need to be learned?
- What attitude needs to be changed?
- How is that reflected in the schedule & program?

EXPECTATIONS

Expectations are based on previous knowledge

Word-of-mouth, Communities of Practice (COP) from ASEAN Member States (AMS)

COST

What are the fixed & variable costs of this project/program for them?

REVENUE

What are the potential financial revenue streams for them?

SATISFACTION

What & How would they tell others?
How would they qualify this experience?

BECOME MEMBER OF Communities of Practice (COP); & Regional Networking; & CERTIFIED GEOSPATIAL-AI INNOVATOR (CGAI²)

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WHAT IS A GEOAI INNOVATOR?

A GeoAI Innovator leverages geospatial data, AI and domain knowledge to solve real-world problems **with impact and integrity**, creating value for people, planet and society— and **turning ideas into scalable, sustainable solutions.**

1. KNOWLEDGE

- **Geospatial Foundations**
  GIS concepts, spatial data models, coordinate systems, map projections
- **GeoAI & Data Science**
  Machine learning, deep learning, statistics, data mining
- **Data & Technologies**
  Remote sensing, GPS/IoT, databases, cloud & big data platforms
- **Programming & Tools**
  Python/R, GIS tools (QGIS, ArcGIS), Google Earth Engine, APIs
- **Domain & Context Knowledge**
  Understanding of sectors, policies, regulations and real-world challenges
- **Ethics, Privacy & Governance**
  Responsible AI, data privacy, security, standards and governance frameworks

2. SKILLS & COMPETENCIES



3. ENTREPRENEURIAL & BUSINESS COMPETENCIES

- **Opportunity Identification**
  Spot real needs and market gaps using geospatial intelligence
- **Value Proposition & Innovation**
  Create unique, user-centered, impactful solutions
- **Business Model & Strategy**
  Design viable business models and go-to-market strategies
- **Financial Literacy**
  Understand costs, pricing, ROI, funding and financial sustainability
- **Stakeholder & Customer Engagement**
  Build networks, partnerships and strong customer relationships
- **Leadership & Execution**
  Lead teams, manage projects, adapt and scale solutions

REAL-WORLD IMPACT

- ✓ Solves real problems
- ✓ Creates social, economic and environmental value
- ✓ Builds a better, smarter and more resilient world

MINDSET OF A GEOAI INNOVATOR

- Curious & Lifelong Learner
- Empathetic & Purpose-Driven
- Ethical & Responsible
- Adaptable & Resilient
- Bold & Impact-Oriented

ULTIMATE GOAL

Transform geospatial data and AI into innovations that create sustainable value and empower communities.

KNOWLEDGE + SKILLS + ENTREPRENEURSHIP + PURPOSE = GEOAI INNOVATOR

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SECTION 1: GEOAI INNOVATION CANVAS

DIMENSION 1 – PAINS

What are the frustrations/problems they want to share related to GeoAI?



WHY IT MATTERS

Understanding these pains helps us design GeoAI solutions that are relevant, practical and impactful for ASEAN.

PAIN / FRUSTRATION	EXAMPLE – TELECOMMUNICATIONS GEOAI RELATED PROBLEM (ASEAN)
Lack of accurate geospatial data	In the Philippines, inconsistent building footprint data leads to inaccurate mobile network coverage maps.
Data is siloed and hard to access	In Indonesia, telecom operators struggle to access government geospatial data in a timely manner.
Limited integration of AI with geospatial data	In Vietnam, AI models are not fully integrated with spatial data, limiting prediction accuracy for network planning.
High cost of data and processing	In Thailand, high-resolution satellite data and processing costs are a barrier for smaller telecom companies.
Shortage of skilled GeoAI talent	In Malaysia, there is a limited pool of professionals skilled in both geospatial analytics and AI.
Privacy and data security concerns	In Singapore, strict data privacy regulations make it challenging to use location data for AI-driven solutions.
Difficulty in real-time decision making	In the Philippines, slow data updates delay real-time network outage detection during disasters.
Lack of standardisation and interoperability	In Cambodia, incompatible data formats and standards make cross-agency data integration difficult.

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GEOAI INNOVATION CANVAS

FROM PAINS TO REAL-PROBLEM USE CASE

IDENTIFIED PAINS

Lack of accurate geospatial data
Inconsistent building footprint data leads to inaccurate mobile network coverage maps.

Data is siloed and hard to access
Telecom operators struggle to access government geospatial data in a timely manner.

Limited integration of AI with geospatial data
AI models are not fully integrated with spatial data, limiting prediction accuracy.

High cost of data and processing
High-resolution satellite data and processing costs are a barrier for smaller companies.

Shortage of skilled GeoAI talent
Limited professionals skilled in geospatial analytics and AI.

Privacy and data security concerns
Strict data privacy regulations make it challenging to use location data for AI-driven solutions.

Difficulty in real-time decision making
Slow data updates delay real-time network outage detection during disasters.

Lack of standardisation and interoperability
Incompatible data formats and standards make cross-agency data integration difficult.

CONVERTED INTO REAL-PROBLEM USE CASE (TELECOMMUNICATIONS & HAZARD RESPONSE)

REAL-WORLD PROBLEM USE CASE

Delayed and Inaccurate Telecom Outage Detection During Disasters in the Philippines

During typhoons and floods, telecom operators face delayed and inaccurate detection of network outages because of:

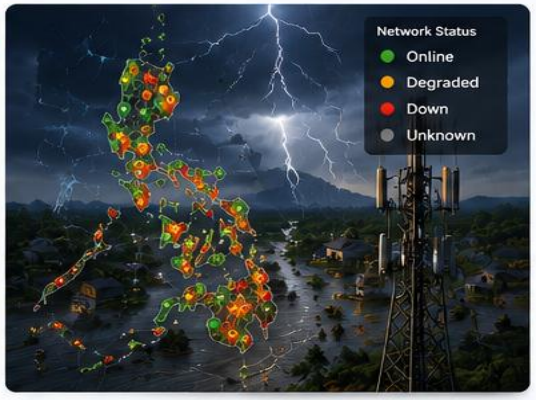
- ✕ Inaccurate and outdated tower and infrastructure locations
- ✕ Siloed and incomplete data from multiple agencies
- ✕ Lack of real-time spatial analytics and AI integration
- ✕ Slow situation awareness and poor damage assessment

THE PAIN IN ACTION

After Typhoon Odette (Rai) in 2021, millions in the Philippines experienced long communication blackouts. Operators struggled to quickly identify which cell towers were up, damaged, or inaccessible due to outdated maps, limited real-time data, and lack of integrated hazard analytics.

REAL NEED

A GeoAI solution that can integrate real-time hazard data, accurate infrastructure data, and AI models to detect outages faster, predict impacts, and prioritise restoration.



IMPACT OF THE PAIN

- Delayed restoration of critical communication services
- Reduced public safety and emergency response effectiveness
- Higher operational cost and revenue loss for telecom operators
- Erosion of customer trust and satisfaction

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GEOAI INNOVATION CANVAS

DIMENSION 2 – SUSTAINABLE COMMITMENT

What sacrifices are you making to sustain and complete the innovation until final Hackathon?



> Innovation requires sacrifice today for a greater impact tomorrow.

SACRIFICE AREA	ONE LINE EXAMPLES OF SACRIFICES
Time Commitment	I sacrifice my free time, weekends, and holidays to work on this GeoAI solution.
Personal & Family Time	I sacrifice time with my family and friends to focus on building and improving the innovation.
Financial Resources	I invest my own money for data access, tools, cloud credits, travel, and other project needs.
Academic / Work Priorities	I postpone or reduce other assignments, research, or work tasks to prioritise this innovation.
Energy & Well-being	I sacrifice sleep, rest, and personal comfort to meet milestones and improve our solution.
Other Opportunities	I turn down other opportunities and activities to stay focused on completing the innovation.
Consistent Effort	I commit to consistent practice, learning, and team collaboration until the final Hackathon.
Long-term Commitment	I stay committed to seeing this innovation through, from ideation to final Hackathon.

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DIMENSION 3

RETURN

What do you expect in return after attending the **bootcamp** and **hackathon**?



What do you expect in return after attending the bootcamp and hackathon?



CERTIFIED

Earn recognition as a **GeoAI Innovator** with industry-relevant certification.



EMPOWERED

Gain the skills, confidence and network to create impact in your community and beyond.



EXPECTATION: | I expect to leave the bootcamp and hackathon **certified and empowered** to turn ideas into real-world solutions.

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DIMENSION 4

GAINS

What do you take away after the Bootcamp & Hackathon ends?

What benefits are you seeking?

RECOGNITION & CREDENTIAL

Earn a recognized certification that validates your GeoAI skills and innovation potential.

SKILLS & KNOWLEDGE

Strengthen practical skills in GeoAI, data analytics, and problem-solving for real-world impact.

INNOVATION & SOLUTION READINESS

Refine your ideas into feasible, scalable solutions with greater clarity and confidence.

NETWORK & COLLABORATION

Build valuable connections with mentors, peers, experts, and potential partners across ASEAN and beyond.

CAREER & OPPORTUNITY

Open doors to internships, career opportunities, and future collaboration in GeoAI and tech ecosystems.

IMPACT & CONTRIBUTION

Contribute to sustainable solutions that create positive impact for your community and the region.

BY THE END OF THE BOOTCAMP & HACKATHON,
I gain the skills, network, confidence, and recognition to innovate and create impact.

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DIMENSION 5

ENTERING BEHAVIOUR

Before this program starts – how would you describe yours and other participants' behaviour?

What are their plausible knowledge, skills and attitudes?



KNOWLEDGE

- Basic understanding of GIS and geospatial concepts.
- Familiar with common data types (spatial and non-spatial) and data sources.
- Some exposure to tools (e.g., QGIS, Google Earth, Excel, or coding basics).
- General awareness of AI/ML and emerging technologies but limited depth.
- Eager to learn but knowledge is mostly theoretical or from past academic experience.

SKILLS

- Can navigate and visualize maps but limited in advanced spatial analysis.
- Basic data handling and analysis (e.g., Excel, basic statistics).
- Some programming exposure (e.g., Python/R) but not confident in applying to real problems.
- Problem-solving skills are developing, especially for complex, real-world challenges.
- Limited experience in data fusion, cloud platforms, or large-scale geospatial workflows.

ATTITUDES

- Motivated and curious to learn new technologies and approaches.
- Open to collaboration and learning from peers and mentors.
- Positive mindset to take on challenges and step out of their comfort zone.
- Interested in using GeoAI to solve real-world problems and create impact.
- Committed to personal growth and contributing to their community or organisation.

IN SUMMARY

Participants are **enthusiastic learners** with foundational knowledge, emerging skills, and a strong **willingness to grow** and collaborate **to create solutions that matter.**

At the start, participants come with **curiosity, potential, and passion** — ready to learn, collaborate and **transform ideas** into impactful GeoAI solutions.

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DIMENSION 6

EXITING BEHAVIOUR

As a result of this program **new skills, attitude and networking** are expected to be acquired.



NEW SKILLS

- Advanced geospatial analysis and visualization using GIS and GeoAI tools.
- Data handling, data fusion, and AI/ML techniques for real-world problem solving.
- Programming confidence (e.g., Python) and ability to build and prototype GeoAI solutions.
- Experience with cloud platforms and modern geospatial workflows.
- Stronger analytical thinking, critical problem-solving and innovation skills.

NEW ATTITUDE

- Confident, motivated and ready to tackle complex challenges.
- A growth mindset with a passion for continuous learning and innovation.
- Collaborative, open-minded and empathetic team player.
- Responsible and ethical use of data and technology for positive impact.
- Resilient, adaptable and committed to turning ideas into sustainable solutions.

ENHANCED NETWORKING

- Stronger connections with peers, mentors, experts and industry professionals across ASEAN and globally.
- Access to a wider community of GeoAI innovators and opportunities for collaboration.
- Long-term relationships that support knowledge sharing, partnerships and co-creation.
- Greater visibility for ideas and solutions through the hackathon ecosystem.

OUTCOME

=

Empowered GeoAI Innovators ready to lead, collaborate and create real-world impact for their communities and the region.

Participants leave the program with **new skills**, **a positive attitude**, and a **strong network** — empowered to innovate, lead and create lasting impact.

SKILLED | EMPOWERED | CONNECTED | IMPACTFUL

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DIMENSION 7

EXPECTATIONS

Expectations are based from previous knowledge:
Word of Mouth, Communities of Practice (COP) and ASEAN Member States (AMS)



1. WORD OF MOUTH

Hearing from peers, alumni and past participants who have attended the bootcamp & hackathon.

We expect:

- High-quality, hands-on learning from experienced trainers and mentors.
- A supportive and inspiring learning environment.
- Practical knowledge that solves real challenges.
- A life-changing experience that builds confidence and opens new opportunities.

2. COMMUNITIES OF PRACTICE (COP)

Insights and experiences shared within GeoAI and related Communities of Practice.

We expect:

- Access to cutting-edge GeoAI tools, data and methodologies.
- Collaboration with experts and innovator networks.
- Guidance to turn ideas into viable, scalable solutions.
- Continued engagement beyond the event through COPs and knowledge sharing.

3. ASEAN MEMBER STATES (AMS)

Expectations shaped by learning needs and development priorities across ASEAN.

We expect:

- Solutions that address real challenges in our countries and region.
- Cross-country collaboration and knowledge exchange.
- Capacity building that strengthens our teams and institutions.
- Contribute to a more innovative, connected and resilient ASEAN.

OVERALL EXPECTATION

An impactful bootcamp & hackathon that empowers us with skills, strengthens networks, and enables us to create meaningful GeoAI solutions for our communities and ASEAN's future.

WE LOOK FORWARD TO:

- Learning from experts and real use cases.
- Building solutions that create real impact.
- Connecting with innovators and potential partners.
- Gaining recognition and advancing our careers.
- Contributing to ASEAN's sustainable and digital future.

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DIMENSION 8

COST

What are their fixed and variable costs that will be incurred?



NOTE

Actual costs may vary depending on location, mode of participation (online or in-person), and individual choices.

FIXED COSTS

Costs that are committed and mostly non-refundable

COST ITEM	DESCRIPTION	PLAUSIBLE COST (USD)
Registration Fee	Covers program participation, materials, platform access, mentoring and event management.	100 – 200
Certification Fee	Cost for certification assessment and issuance (if applicable).	50 – 100
Event Kit & Materials	Includes bootcamp kit, handbook, stationery, merchandise, etc.	20 – 40
Digital Platform Access	Access to learning platform, datasets, tools and cloud resources during the program.	20 – 50
Mentorship & Expert Sessions	Facilitated sessions, expert talks and mentoring support.	10 – 30
ESTIMATED TOTAL FIXED COST		200 – 420

VARIABLE COSTS

Costs that vary based on individual choices and usage

COST ITEM	DESCRIPTION	PLAUSIBLE COST (USD)
Travel (If applicable)	Flights, train or bus fares to attend in-person bootcamp or final presentation.	100 – 600
Accommodation (If applicable)	Hotel/hostel stays during the event days.	50 – 200/night
Meals & Refreshments (If not provided)	Daily meals, snacks, coffee, beverages.	10 – 30/day
Local Transportation	Airport transfers, public transport, ride-hailing, parking, etc.	10 – 50/day
Internet & Communication	Mobile data, local SIM card, or roaming charges.	5 – 20
Incidentals & Personal Expenses	Printing, team supplies, souvenirs, personal needs, contingencies.	20 – 50
ESTIMATED TOTAL VARIABLE COST (PER PARTICIPANT)		195 – 1,100+

PLANNING AHEAD HELPS!

We encourage participants to plan their budget early, **seek support or sponsorship if needed**, and make the most of this transformative learning and innovation experience.

SUMMARY

Total estimated cost (per participant)

- Without travel & accommodation: USD 200 – 420
- With travel & accommodation: USD 395 – 1,520+

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DIMENSION 9

REVENUE

What are their fixed potential financial revenue streams that is plausible for them?



FIXED POTENTIAL FINANCIAL REVENUE STREAMS

Stable and recurring income opportunities from applying skills, solutions and networks gained

1 EMPLOYMENT INCREASE / SALARY GROWTH

Apply new GeoAI skills to qualify for higher-value roles or promotions.

How you earn:

Higher salary, allowances or performance bonuses.

Example: RM 3,000 – RM 10,000+ more per year

2 CONSULTING & FREELANCE PROJECTS

Offer GeoAI analysis, mapping, data insights and solution development to clients.

How you earn:

Project-based fees or retainer arrangements.

Example: RM 2,000 – RM 20,000+ per project

3 DIGITAL PRODUCTS & TOOLS

Create and sell digital assets such as datasets, dashboards, templates, or AI models.

How you earn:

One-time sales or subscription fees.

Example: RM 500 – RM 5,000+ per month

4 TRAINING & WORKSHOPS

Deliver training, bootcamps, or corporate workshops on GeoAI, GIS and data analytics.

How you earn:

Facilitation fees or institutional contracts.

Example: RM 3,000 – RM 15,000+ per session

5 SOLUTION LICENSING

License your developed solutions, apps or algorithms to organisations or government.

How you earn:

License fees or usage-based pricing.

Example: RM 5,000 – RM 50,000+ per license

6 GRANTS, FUNDING & COMPETITIONS

Win grants, innovation funds or prizes for your solutions and research outcomes.

How you earn:

Grant awards, innovation funds or prize money.

Example: RM 5,000 – RM 100,000+ per award

KEY TAKEAWAY

The skills, networks and solutions you gain from this Bootcamp & Hackathon can open multiple, stable revenue opportunities that grow over time.

REVENUE IMPACT

From small beginnings to long-term financial growth and impact.

TURN SKILLS INTO SUSTAINABLE INCOME

GeoAI is in high demand across industries and governments in ASEAN. Your expertise today can become your financial advantage tomorrow.

High demand skills → Strong networks → Valuable solutions → Sustainable revenue

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DIMENSION 10

SATISFACTION

What and how would you tell others about this experience?



WHAT I WILL TELL OTHERS

- An inspiring and practical learning journey.
- Hands-on experience with real-world GeoAI challenges.
- Expert mentors and supportive community.
- A life-changing experience that builds skills, confidence and purpose.
- Well-organized, engaging and impactful Bootcamp & Hackathon.

HOW I WILL SHARE MY EXPERIENCE

- Share on social media with posts, photos and achievements.
- Recommend to friends, classmates and colleagues.
- Present our solutions and learnings in school, workplace or community.
- Stay active in Communities of Practice (COP) and continue collaborating.
- Be an ambassador and inspire others to join future programs.

THE IMPACT I WILL HIGHLIGHT

- New skills in GeoAI, data and technology.
- Real solutions that address local and regional challenges.
- Global exposure and cross-country collaboration within ASEAN.
- Opportunities for innovation, career growth and entrepreneurship.
- Recognition and certification that validates my journey.

HOW THIS EXPERIENCE MAKES ME FEEL

- Empowered to create impact through GeoAI solutions.
- Confident in my abilities and future.
- Connected to a strong network of innovators and mentors.
- Proud to contribute to my community, country and ASEAN.
- Motivated to keep learning, innovating and leading.

YOU CAN ACQUIRE RECOGNITION THROUGH THREE LEVELS:

LOWEST LEVEL

GeoAIExplorer (GAIEx)

For participants who successfully complete the Bootcamp and Hackathon.

- Demonstrates foundational understanding of GeoAI.
- Actively participates and completes challenges.
- Ready to continue learning and growing.

RECOGNIZED AS A PROMISING GEOAI EXPLORER

MID LEVEL

Associate GeoAI Innovator (AGAI)

For participants who show strong application skills and deliver quality solutions.

- Applies GeoAI tools and techniques effectively.
- Collaborates well and shows problem-solving ability.
- Delivers innovative solutions with real-world relevance.

RECOGNIZED AS AN ASSOCIATE GEOAI INNOVATOR

HIGHEST LEVEL

Certified GeoAI Innovator (CGAI)

For top performers who demonstrate excellence, impact and leadership.

- Delivers outstanding, impactful GeoAI solutions.
- Shows leadership, innovation and influence.
- Committed to creating sustainable and meaningful impact.

RECOGNIZED AS A CERTIFIED GEOAI INNOVATOR

MY SATISFACTION. MY STORY. MY IMPACT.

I will proudly share my journey, inspire others, and continue to grow as a GeoAI innovator for a better future.

INSPIRE OTHERS | BUILD NETWORKS | CREATE SOLUTIONS | DRIVE IMPACT

Learn. Innovate. Impact.

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ASEAN GEOAI FUSION BOOTCAMP & HACKATHON

FROM LEARNING JOURNEY TO REAL-WORLD IMPACT

ENTERING BEHAVIOUR

Where you start

Knowledge
- Basic understanding of GIS, data & AI
- Some exposure to tools or programming
- Awareness of real-world challenges

Skills
- Foundational technical skills
- Problem-solving at a basic level
- Limited experience in data analysis & modelling

Attitudes
- Curious and eager to learn
- Open to collaboration
- Motivated to make an impact

I am here to learn, collaborate and grow.

4 SUB DIMENSIONS THAT TRANSFORM YOUR JOURNEY

1 EXPERIENCE JOURNEY

What chronological touch points should they experience?

1. **Discover** – Learn about the program, objectives and opportunities
2. **Onboard** – Orientation, setup and community introduction
3. **Learn** – Foundational & advanced learning sessions, mentorship
4. **Collaborate** – Team formation, ideation and solution design
5. **Build** – Prototype development and testing
6. **Pitch & Validate** – Present solutions, receive feedback from judges & mentors
7. **Celebrate & Connect** – Recognition, networking and next steps

A guided journey from learning to impact.

2 JOBS TO BE DONE

What are your to solve list?

MY TO SOLVE LIST

- Understand the real problems in my community/country.
- Collect, integrate and analyze geospatial data.
- Apply AI/ML to extract insights and predict outcomes.
- Design innovative, scalable GeoAI solutions.
- Validate solutions with users and stakeholders.
- Communicate impact and recommend for adoption.
- Contribute to a better and more resilient ASEAN.

I come to solve real problems with data, AI and collaboration.

3 PROMISE

How does this promise alleviate pains & create gains?

ALLEVIATES PAINS

- Overcome lack of skills and confidence.
- Access quality data, tools and expert guidance.
- Reduce time and cost of trial and error.
- Navigate complexity of real-world problems.
- Feel less alone – be part of a supportive community.

CREATES GAINS

- Gain new skills and practical experience.
- Build impactful solutions that create value.
- Expand network across ASEAN and beyond.
- Earn recognition and credentials.
- Increase career and innovation opportunities.

We remove barriers so you can gain skills, solutions and success.

4 INSTRUCTIONAL DESIGN

How can you best learn? What attitudes need to change? How can you best solve your real problem?

HOW YOU LEARN BEST

- Hands-on, project-based learning
- Real-world case studies & datasets
- Collaborative teamwork & peer learning
- Mentorship from experts & practitioners
- Iterative feedback & continuous improvement

ATTITUDES TO STRENGTHEN

- From "I don't know" to "I am curious"
- From "I work alone" to "We collaborate"
- From "Problems are big" to "I can solve them"
- From "Avoid risk" to "Embrace innovation"
- From "Learning for me" to "Impact for others"

HOW YOU SOLVE YOUR REAL PROBLEM

- Understand → Empathize → Define
- Collect → Analyze → Generate Insights
- Ideate → Prototype → Test
- Validate → Refine → Implement
- Scale → Share → Sustain Impact

Learn with purpose. Change mindset. Solve with impact.

EXITING BEHAVIOUR

Where you finish

Knowledge
- Advanced GeoAI knowledge
- Deeper understanding of data, tools & AI
- Real-world application expertise

Skills
- Strong analytical & technical skills
- Solution design & prototyping experience
- Confident communicator and problem solver

Attitudes
- Growth mindset
- Collaboration & leadership
- Impact-driven and future-ready

I am empowered to lead, innovate and create impact.

ONE JOURNEY. FOUR PILLARS. TRANSFORMING LEARNERS INTO GEOAI INNOVATORS.

FOUNDATIONAL KNOWLEDGE IN GEOAI

FD1001:CREATING YOUR PROBLEM-DATA-GEOAI SOLUTION CANVAS BASED ON USE-CASES

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ASEAN GEOAI FUSION 2026

PROBLEM-DATA-GEOAI-SOLUTION (PDGS) CANVAS: D11

ASEAN GeoAI FUSION

PROBLEM–DATA–GeoAI SOLUTION CANVAS

From Real-World Problems to Intelligent Geospatial Solutions

Project Name .....

Hackathon Group .....

○ Problem ○ Data ○ GeoAI ○ Solution ○ Impact

1 PROBLEM STATEMENT

Clearly define the real-world problem worth solving.

- What is the problem?
- Who experiences it?
- Why is it important?
- What is the current impact?
- What are the success criteria?

OUTPUTS
- One-sentence problem statement
- Target beneficiaries
- Problem owner
- Success criteria

2 STAKEHOLDERS & USERS

Identify everyone who influences or benefits from the solution.

- Who are the users?
- Who makes decisions?
- Who owns the data?
- Who benefits?
- Who may resist adoption?

OUTPUTS
- Stakeholder map
- User personas
- Stakeholder priorities

3 SPATIAL CONTEXT

Understand the geographic scope and spatial characteristics of the problem.

- Where does it occur?
- What is the geographic scale?
- What spatial relationships are important?
- What are the geographic boundaries?

OUTPUTS
- Area of Interest (AOI)
- Maps
- Geographic scope
- Key spatial characteristics

4 DATA REQUIREMENTS

Identify the right data needed to understand and solve the problem.

Spatial Data
Imagery, GIS layers, DEM, roads, buildings, land use, POIs, etc.

Non-Spatial Data
Sensor, weather, energy, population, mobility, economic, survey, etc.

- Who owns the data?
- What is the quality?
- How often is it updated?
- Are there data gaps?

OUTPUTS
- Data inventory
- Data sources
- Data quality assessment
- Data gap analysis

5 GeoAI INTELLIGENCE

Determine what intelligence GeoAI should generate.

Possible Tasks
- ✓ Classification
- ✓ Prediction
- ✓ Detection
- ✓ Segmentation
- ✓ Change Detection
- ✓ Forecasting
- ✓ Optimization
- ✓ Recommendation

- What decision will AI support?
- What insight should AI generate?
- Why is GeoAI needed (vs. traditional GIS)?

OUTPUTS
- AI objective
- GeoAI workflow
- Expected intelligence outputs

6 GeoAI SOLUTION DESIGN

Translate intelligence into a usable solution for users.

- What will users interact with?
- What features and functions are needed?
- What is the user journey?

OUTPUTS
- Solution concept
- User journey
- Functional modules
- High-level architecture / prototype

7 TECHNOLOGY STACK

Select the technologies and platforms to build the solution.

GIS	ArcGIS, QGIS, etc.
Cloud	GEE, AWS, Azure, GCP, etc.
AI / ML	TensorFlow, PyTorch, Scikit-learn, etc.
Programming	Python, JavaScript, R, etc.
Database	PostgreSQL/PostGIS, etc.
Visualization	Power BI, Dash, Streamlit, etc.

OUTPUTS
- Technology architecture
- Software stack
- Infrastructure requirements

8 ETHICS, PRIVACY & SUSTAINABILITY

Ensure responsible, ethical and sustainable GeoAI.

- Are there privacy concerns?
- Is the solution fair and unbiased?
- Is it explainable and transparent?
- Is it environmentally sustainable?
- Does it comply with regulations?

OUTPUTS
- Ethics checklist
- Risk assessment & mitigation plan
- Sustainability strategy
- Governance requirements

9 VALUE & IMPACT

Define the value the solution will create and how it will be measured.

Economic
Cost savings, ROI, growth

Social
Better services, safety, inclusion

Environmental
Emissions reduction, resilience

Operational
Efficiency, automation, speed

KPIs (Examples)
- ✓ Cost savings
- ✓ % Improvement
- ✓ No. of people impacted
- ✓ Time saved
- ✓ CO_2 reduction
- ✓ Accuracy
- ✓ Utilization rate
- ✓ Adoption rate

OUTPUTS
- Impact statement
- KPI scorecard
- Benefits map / value proposition

10 IMPLEMENTATION ROADMAP

Plan how the solution will be implemented and scaled.

1. **Prototype** — Concept & initial model
2. **Pilot** — Test in real-world setting
3. **Deployment** — Full-scale rollout
4. **Scale** — Expand to more areas/users
5. **Sustain** — Monitor, improve, maintain

OUTPUTS
- Roadmap & timeline
- Resources & budget
- Team & partners
- Risks & mitigation
- KPIs for each phase

THE JOURNEY

From Problem to Impact

PROBLEM — Understand the real challenge

DATA — Collect the right data

GeoAI — Generate intelligence & insights

SOLUTION — Deliver usable solutions for decisions

IMPACT — Create measurable value & sustainable change

Better Decisions
Smarter Communities
Sustainable Future

ASEAN GeoAI FUSION

PROBLEM–DATA–GeoAI SOLUTION CANVAS

From Real-World Problems to Intelligent Geospatial Solutions

Problem · Data · GeoAI · Solution · Impact

1 PROBLEM STATEMENT

Clearly define the real-world problem worth solving.

- What is the problem?
- Who experiences it?
- Why is it important?
- What is the current impact?
- What are the success criteria?

OUTPUTS
- One-sentence problem statement
- Target beneficiaries
- Problem owner
- Success criteria

2 STAKEHOLDERS & USERS

Identify everyone who influences or benefits from the solution.

- Who are the users?
- Who makes decisions?
- Who owns the data?
- Who benefits?
- Who may resist adoption?

OUTPUTS
- Stakeholder map
- User personas
- Stakeholder priorities

3 SPATIAL CONTEXT

Understand the geographic scope and spatial characteristics of the problem.



- Where does it occur?
- What is the geographic scale?
- What spatial relationships are important?
- What are the geographic boundaries?

OUTPUTS
- Area of Interest (AOI)
- Maps
- Geographic scope
- Key spatial characteristics

4 DATA REQUIREMENTS

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- ✓ Recommendation

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- GeoAI workflow
- Expected intelligence outputs

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- Software stack
- Infrastructure requirements

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- **Operational** — Efficiency, automation, speed

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- ✓ Co_2 reduction
- ✓ Accuracy
- ✓ Utilization rate
- ✓ Adoption rate

OUTPUTS
- Impact statement
- KPI scorecard
- Benefits map / value proposition

10 IMPLEMENTATION ROADMAP

Plan how the solution will be implemented and scaled.



OUTPUTS
- Roadmap & timeline
- Resources & budget
- Team & partners
- Team & partners
- Risks & mitigation

THE JOURNEY

From Problem to Impact

PROBLEM — Understand the real challenge

DATA — Collect the right data

GeoAI — Generate intelligence & insights

SOLUTION — Deliver usable solutions for decisions

IMPACT — Create measurable value & sustainable change

Better Decisions
Smarter Communities
Sustainable Future

ASEAN GeoAI FUSION

USE CASE 1

GeoAI Use Case

Telecommunications & Hazard Response

Keeping Communities Connected When Disasters Strike

Use Case: Intelligent Network Resilience and Rapid Response Platform

Detect Risk | Predict Impact | Prioritise Action | Restore Connectivity

Slide 1

1 PROBLEM STATEMENT

Real-world problem worth solving.

- Natural hazards such as floods, storms, landslides and earthquakes often damage telecommunication infrastructure.
- Network outages disrupt emergency communication and delay rescue operations.
- Operators lack a real-time, predictive view of risk, impact and recovery priorities.
- This leads to slow response, higher operational cost, and impacts to lives and livelihoods.

IMPACT

- Communication blackouts
- Delayed emergency response
- Safety risks for communities
- Revenue & reputation loss

SUCCESS CRITERIA

- Reduce network outage duration
- Improve response time
- Optimise resource deployment
- Increase network resilience
- Protect lives and livelihoods

Slide 2

2 STAKEHOLDERS

Key stakeholders and their roles.



- **Telecom Operators** — Network data, operations, field teams
- **Government Agencies** — Disaster management, policy, coordination
- **Technology Partners** — Cloud, AI/ML, GIS, IoT
- **Emergency Services** — Rescue, medical, public safety
- **Communities** — Citizens, users, feedback
- **Local Authorities** — Local information, permits, ground support

Slide 3

3 SPATIAL CONTEXT

Where does the problem occur?

- **Area of Interest (AOI):** High-risk region / province / country
- **Geographic Scale:** National to local (multi-scale)

Key Spatial Considerations

- Hazard-prone areas
- Population distribution
- Critical sites & infrastructure
- Transport accessibility
- Terrain & elevation
- Land use / land cover



Slide 4

4 DATA REQUIREMENTS

The right data before selecting AI methods.

SPATIAL DATA

- Hazard layers (flood, storm, landslide, earthquake)
- Telecom tower & site locations
- Fiber routes & network topology
- Road network
- Elevation / DEM
- Land use / land cover
- Population density
- Weather radar & satellite imagery

NON-SPATIAL DATA

- Network performance data (alarms, downtime)
- Power availability
- Equipment inventory
- Maintenance & ticket data
- Historical incident reports
- Field team status & locations
- Customer complaints / CDR data

DATA SOURCES (EXAMPLES)

MCMC | METMalaysia | JUPEM | OpenStreetMap | Sentinel Hub | Operators' Data (OSS/BSS/EMS)

Slide 5

5 GeoAI INTELLIGENCE

Intelligence GeoAI should generate.

GEOAI TASKS

- Hazard Impact Prediction (where & what will be affected)
- Network Outage Prediction (sites at risk)
- Vulnerability Assessment (criticality & exposure)
- Recovery Time Estimation (how long to restore)
- Resource Optimisation (best allocation of teams/assets)

GeoAI

GEOAI OUTPUTS

- Risk / Impact Heatmap
- Outage Probability Map
- Critical Sites Dashboard
- Recovery Priority List
- Recommended Actions & Optimal Routes

AI + GEOSPATIAL ANALYTICS = SMARTER, FASTER DECISIONS

Slide 6

6 GeoAI SOLUTION DESIGN

Translate intelligence into a usable solution.

SOLUTION NAME

Telecom Resilience & Response (TRR) Platform

Risk & Impact Dashboard | Outage Prediction | Resource Optimisation | Field Operations Mobile App | Alerts & Notifications



Slide 7

7 TECHNOLOGY STACK

Technologies appropriate to the project.

- **GIS** — ArcGIS Pro, QGIS
- **Cloud** — AWS, Azure, Google Cloud
- **AI / Machine Learning** — Python, TensorFlow, PyTorch
- **Data Platform** — PostgreSQL / PostGIS, BigQuery
- **Streaming & IoT** — Kafka, MQTT, IoT Sensors
- **Dashboard & Apps** — Power BI, ArcGIS Dashboard, Flutter



Slide 8

8 ETHICS, PRIVACY & SUSTAINABILITY

Responsible, ethical and sustainable GeoAI.

- **PRIVACY** — Anonymise CDR and user data.
- **ETHICS** — Ensure fair access to communication services.
- **SECURITY** — Protect sensitive infrastructure and data.
- **SUSTAINABILITY** — Improve resilience, reduce energy use, support green recovery.



11 SUSTAINABLE CITIES AND COMMUNITIES | 13 CLIMATE ACTION | 9 INDUSTRY, INNOVATION AND INFRASTRUCTURE | 17 PARTNERSHIPS FOR THE GOALS

9 VALUE & IMPACT

Value the solution creates and how it will be measured.

- **HUMAN IMPACT** — Faster communication restoration, saves lives.
- **OPERATIONAL IMPACT** — Lower downtime, optimised operations, cost savings.
- **SOCIAL IMPACT** — Improved public safety and trust.
- **ENVIRONMENTAL IMPACT** — Smarter resource use, lower carbon footprint.

KPIs (Examples)

- 30–50% Reduction in outage duration
- 25–40% Improvement in response time
- 20–30% Better accuracy in outage prediction
- Improved customer satisfaction (NPS)

10 IMPLEMENTATION ROADMAP

From pilot to full-scale deployment.



1. **Data Collection** (0–3 Months) — Integrate network, hazard & external datasets
2. **Model Development** (3–6 Months) — Build predictive models & risk layers
3. **Pilot Deployment** (6–9 Months) — Test in high-risk area, refine models
4. **Scale & Integration** (9–15 Months) — Integrate with NOC, field ops & partners
5. **Continuous Improvement** (Ongoing) — Monitor, learn, and enhance continuously

11 FINAL PITCH

Summarise the solution in one page.

- **THE PROBLEM** — Natural hazards disrupt telecom networks, delaying communication and saving lives.
- **THE DATA** — Integrate hazard, network, operational and contextual data.
- **THE GeoAI SOLUTION** — Predict impact, prioritise actions and optimise recovery with GeoAI.
- **THE IMPACT** — Faster response, resilient networks, protected communities.
- **LOW-CARBON FUTURE** — Smarter, more sustainable and resilient digital infrastructure.



ASEAN GEOAI FUSION 2026

GEOAI AROUND US

FOUNDATIONAL KNOWLEDGE IN GEOAI



FOUNDATIONAL KNOWLEDGE IN GEOAI

1

PROBLEM STATEMENT

Real world problem worth solving



Natural hazards such as floods , storms, landslides, earthquakes often damage telecommunications infrastructure



Network outages disrupt emergency communication & delay rescue operation



Operators lack real-time, predictive view of risk, impact & recovery priorities



This leads to slow response, higher operational costs, & impact to lives & livelihoods

Success Criteria

Reduce network outage duration & Increase network resilience

IMPACT



Communications Blackouts



Delayed emergency response



Safety risks for communications



Revenue & reputation loss

FOUNDATIONAL KNOWLEDGE IN GEOAI

2

STAKEHOLDERS

Key Stakeholders & their Roles



KEY STAKEOLDERS


Telecom Operators


Technology Partners



Government Agencies



Emergency Services


Local Authorities


Communities

FOUNDATIONAL KNOWLEDGE IN GEOAI

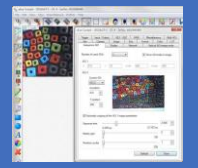
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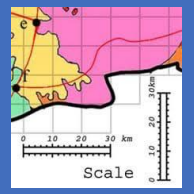
SPATIAL CONTEXT

Where does the problem Occur?



SPATIAL CONTEXT





AOI Areas of Interest
High risk
region/province/country
Geographic Scale
National/Regional /local

KEY SPATIAL CONSIDERATIONS

- Hazard-prone areas
- Population distribution
- Critical sites & infrastructure
- Transport accessibility
- Terrain & Elevation
- Land use/Land cover

FOUNDATIONAL KNOWLEDGE IN GEOAI

4

DATA REQUIREMENTS

The right data before selecting AI Methods & Tools

SPATIAL DATA

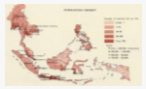

Hazard layers (floods, storms, landslides & earthquakes)

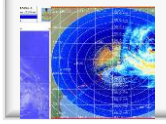

Telecom towers & Site locations


Fibre route & Network topology


Road Network


Elevation/Digital Elevation Model (DEM)


Population Density


Weather radar & Satellite Imagery

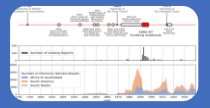
NON-SPATIAL DATA


Network performance data (alarms, downtime)


Power Availability


Equipment Inventory


Maintenance & Ticket Data


Historical Incident Report


Field Team Status & Locations


Customer complaints & CDR Data

FOUNDATIONAL KNOWLEDGE IN GEOAI

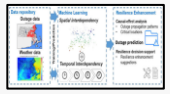
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GEOAI INTELLIGENCE

Intelligence GeoAI should generate

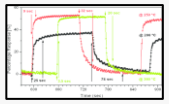
GEOAI TASKS


Hazard impact prediction


Network outage prediction (Sites at risks)



Vulnerability Assessment (criticality & exposure)

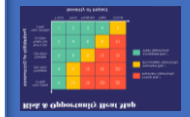

Recovery time estimation (how long to restore)

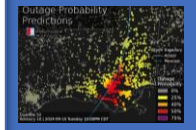


Resource Optimisation (Best allocation of Teams & Assets)



GEOAI OUTPUTS


Risk & Impact Heatmap



Outage probability Map


Critical sites dashboard


Recovery priority list


Recommended Actions & Optimal routes

FOUNDATIONAL KNOWLEDGE IN GEOAI

6

GEOAI SOLUTION DESIGN

Translate Intelligence into a usable solution



KEY FEATURES

- Risk & Impact dashboard
- Outage Prediction
- Resource Optimisation
- Alerts & Notifications

FOUNDATIONAL KNOWLEDGE IN GEOAI

7

TECHNOLOGY STACK

Technologies appropriate to the solution/project



TECHNOLOGIES

- GIS
- Cloud
- AI, ML
- Data platform
- Streaming & IoT
- Dashboard & Apps

FOUNDATIONAL KNOWLEDGE IN GEOAI

8

ETHICS, PRIVACY & SUSTAINABILITY

Responsible, Ethical & Sustainable GeoAI

PRIVACY
Anonymise CDR and user data.

ETHICS
Ensure fair access to communication services.

SECURITY
Protect sensitive infrastructure and data.

SUSTAINABILITY
Improve resilience, reduce energy use, support green recovery.





SUSTAINABLE DEVELOPMENT GOALS (SDGs)

- 11 SUSTAINABLE CITIES AND COMMUNITIES — Sustainable cities & Communities
- 13 CLIMATE ACTION — Climate Action
- 9 INDUSTRY, INNOVATION AND INFRASTRUCTURE — Industry Innovation & Infrastructure
- 17 PARTNERSHIPS FOR THE GOALS — Partnerships for Goals

FOUNDATIONAL KNOWLEDGE IN GEOAI

9

VALUE & IMPACT

Value the solution creates & how it will be measured

HUMAN IMPACT
Faster communication restoration, saves lives.

OPERATIONAL IMPACT
Lower downtime, optimised operations, cost savings.

SOCIAL IMPACT
Improved public safety and trust.

ENVIRONMENTAL IMPACT
Smarter resource use, lower carbon footprint.

KPIs (Examples)

- 30–50% Reduction in outage duration
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- Improved customer satisfaction (NPS)



FOUNDATIONAL KNOWLEDGE IN GEOAI

10

IMPLEMENTATION ROADMAP

From pilot to full scale deployment



PHASES

Phase 1: Data Collection

Phase 2: Model Development

Phase 3: Pilot Deployment

Phase 4: Scale & Integration

Phase 5 : Continuous Improvement (Ongoing)

FOUNDATIONAL KNOWLEDGE IN GEOAI

11

FINAL PITCH

Summarise the Solution in one page

THE PROBLEM
Natural hazards disrupt telecom networks, delaying communication and saving lives.

THE DATA
Integrate hazard, network, operational and contextual data.

THE GeoAI SOLUTION
Predict impact, prioritise actions and optimise recovery with GeoAI.

THE IMPACT
Faster response, resilient networks, protected communities.

LOW-CARBON FUTURE
Smarter, more sustainable and resilient digital infrastructure.



THE PROBLEM: Natural hazards disrupts telecom networks, delaying communications & saving lives .

THE DATA: Integrate hazard, network, operational and contextual data.

THE GEOAI SOLUTION: Predict impact, prioritise actions, & optimise recovery with GeoAI

THE IMPACT: Faster response, resilient networks, protected communities.

LOW CARBON FUTURE: Smarter, more sustainable & resilient digital infrastructure

ASEAN GeoAI FUSION

USE CASE 2

INTRODUCTION

Smart Energy Planning for EV Charging Stations in Kuala Lumpur

A GeoAI Case Study

This case study demonstrates how GoeAI can help plan optimal EV charging station locations, improve accessibility, maximise utilisation, and accelerate a low-carbon future.

1 PROBLEM STATEMENT

EV adoption in Malaysia is increasing rapidly, but charging infrastructure is unevenly distributed.

THE CHALLENGE

- Long waiting times
- Range anxiety
- Low EV adoption
- Traffic searching for chargers
- Unequal access across areas

SUCCESS CRITERIA

- Reduce average travel distance to a charger
- Improve charging accessibility
- Optimise utilisation
- Support sustainable urban mobility

2 STAKEHOLDERS



3 SPATIAL CONTEXT

Area of Interest:
Kuala Lumpur

Geographic Scale:
City

Key Spatial Considerations

- Residential areas
- Commercial centres
- Highway exits
- Shopping malls
- Universities
- Tourist attractions
- Existing charging stations
- Electrical substations
- Traffic density zones



4 DATA REQUIREMENTS

SPATIAL DATA

- Road network
- Existing EV charging stations
- Land use
- Building footprints
- Parking locations
- Electrical substations
- Public transport stations
- Population density
- Traffic density
- Satellite imagery

NON-SPATIAL DATA

- EV registrations
- Electricity demand
- Charging utilisation
- Weather
- Income levels
- Vehicle ownership
- Electricity tariffs

DATA SOURCES

JUPEM · PLANMalaysia · TNB · MCMC · DOSM · OpenStreetMap

5 GeoAI INTELLIGENCE

AI OBJECTIVE

Predict the best future charging locations.

GeoAI TASKS

- Demand Prediction
- Spatial Clustering
- Suitability Analysis
- Accessibility Modelling
- Grid Capacity Prediction
- Optimisation



GooAI OUTPUTS

- Future charging demand map
- Priority investment zones
- Recommended charging sites
- Expected utilisation
- Grid impact assessment

Leveraging AI + Geospatial Analytics for Smarter Decisions

6 GeoAI SOLUTION DESIGN

Proposed Solution:

GeoChargeAI

An intelligent planning platform for data-driven EV charging infrastructure.

Key Modules

- Interactive GIS Dashboard
- Charging Demand Prediction
- Site Recommendation Engine
- Grid Capacity Monitor
- Carbon Savings Calculator
- Investment Prioritisation



Simulate & Decide:
"What happens if we build 50 new chargers here?"

7 TECHNOLOGY STACK

- **GIS** — ArcGIS Pro, QGIS
- **Cloud** — Google Earth Engine, Azure
- **Machine Learning** — TensorFlow, XGBoost
- **Programming** — Python
- **Database** — PostgreSQL / PostGIS
- **Dashboard** — Power BI, ArcGIS Dashboard



8 ETHICS, PRIVACY & SUSTAINABILITY

PRIVACY
No personal driving data.
Use aggregated mobility data.

ETHICS
Avoid favouring only affluent neighbourhoods.
Ensure equitable access.

SUSTAINABILITY
Support carbon reduction, clean transportation, and SDG 11, SDG 13 (Climate Action).





9 VALUE & IMPACT

ECONOMIC
Better investment decisions, lower costs, higher utilisation.

SOCIAL
Reduced range anxiety, improved accessibility and public convenience.

ENVIRONMENTAL
Increased EV adoption, reduced emissions, cleaner air.

OPERATIONAL
Optimised charging network, improved grid planning.

KPIs (Examples)

- 30% reduction in average travel distance to a charger
- 20% increase in charger utilisation
- 25% reduction in charging wait times
- 15% increase in EV adoption in target areas
- Annual CO_2 emissions avoided

10 IMPLEMENTATION ROADMAP



1. **Data Collection (Months 1–2)**
   - Gather GIS, traffic, energy data
   - Validate data quality
2. **AI Model Development (Months 3–5)**
   - Train demand prediction models
   - Conduct spatial suitability analysis
3. **Pilot (Months 6–8)**
   - Test in selected districts
   - Evaluate predictions
4. **Deployment (Months 9–12)**
   - Roll out city-wide
   - Integrate with planning systems
5. **Continuous Improvement**
   - Update models with new data
   - Expand to other Malaysian cities

11 FINAL PITCH

THE PROBLEM
Uneven distribution of EV charging stations is limiting EV adoption and creating accessibility issues.

THE DATA
Integrate geospatial, traffic, energy, demographic, and EV usage data.

THE GeoAI SOLUTION
GeoChargeAI predicts demand, identifies optimal charging locations, and assesses grid impacts using AI and spatial analytics.

THE IMPACT
More accessible charging infrastructure, higher EV adoption, improved investment efficiency, reduced emissions, and better urban energy planning.

LOW-CARBON FUTURE
Smarter cities • Cleaner energy • Better lives for all

FOUNDATIONAL KNOWLEDGE IN GEOAI



FOUNDATIONAL KNOWLEDGE IN GEOAI

1

PROBLEM STATEMENT

Real World Problem Worth Solving



EV adoption in Malaysia is increasing rapidly, but charging infrastructure is unevenly distributed

Many areas, especially outside major cities, lack sufficient charging stations.



Drivers face long waiting times, range anxiety & extra traffic while searching for chargers

This leads to frustration, wasted time and higher travel costs.



Planners lack a data-driven view of demand, accessibility & grid capacity

Decisions are often based on limited insights, leading to suboptimal infrastructure planning.



This leads to low EV adoption & unequal access to charging across areas

Slows down the transition to sustainable mobility and increases carbon emissions.

SUCCESS CRITERIA

- Reduce travel distance
- Improve accessibility
- Optimise utilisation

IMPACT



Uneven Charging Access

Many areas lack chargers, creating access gaps.



Long waiting times & range anxiety

Drivers face delays, traffic and uncertainty.



Slower EV adoption

Poor infrastructure planning limits market growth.



Congestion from charger searching

Extra traffic, emissions and wasted resources.



FOUNDATIONAL KNOWLEDGE IN GEOAI

2

STAKEHOLDERS

Key Stakeholders & Their Roles



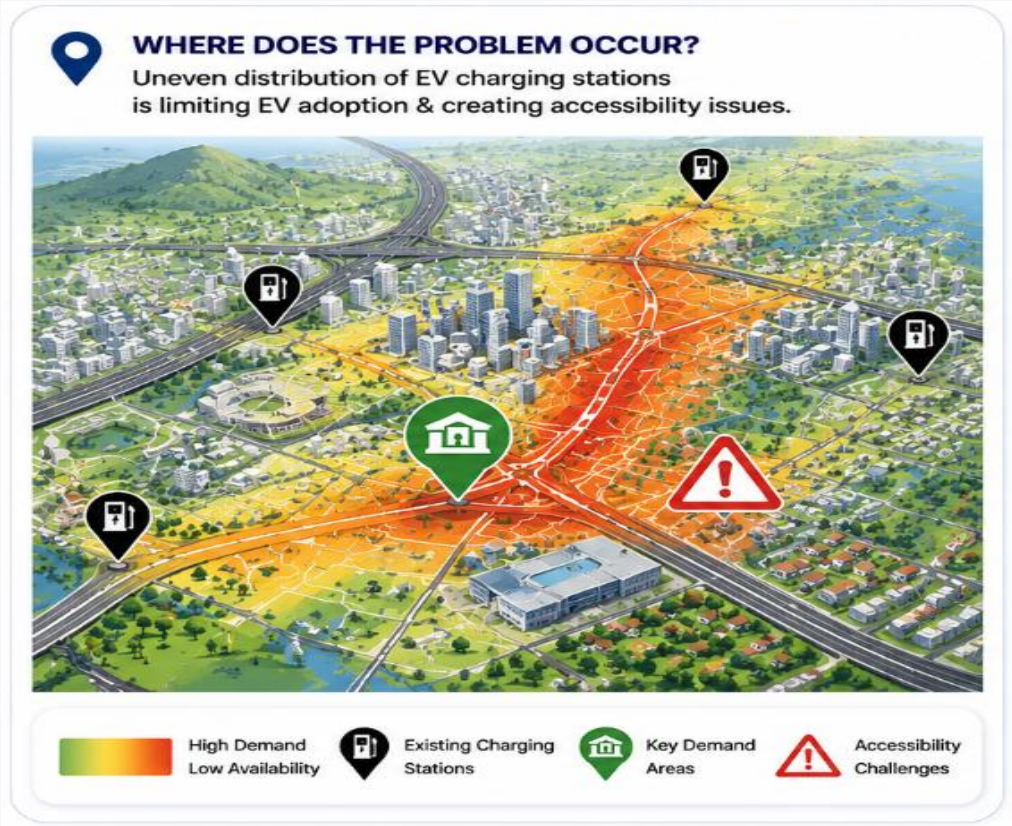


FOUNDATIONAL KNOWLEDGE IN GEOAI

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SPATIAL CONTEXT

Where does the problem Occur?





FOUNDATIONAL KNOWLEDGE IN GEOAI

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DATA REQUIREMENTS

The Right Data Before Selecting AI Methods & Tools

SPATIAL DATA



Road network & traffic density
Road connectivity, speed, congestion levels, and traffic volume.



Existing EV charging stations
Locations, types, connectors, capacity, operator information.



Land use & building footprints
Residential, commercial, industrial zones and building outlines.



Parking locations
On-street and off-street parking availability and capacity.



Electrical substations
Substation locations and capacity for power availability.



Public transport stations & population density
Transit access points and population distribution.



Satellite imagery
High-resolution imagery for environment, land cover and change detection.

NON-SPATIAL DATA

EV registrations
Number of EVs, vehicle types, registration trends over time.

Electricity demand
Historical & forecasted electricity demand by region and time.

Charging utilisation
Usage patterns, peak hours, dwell time, and station occupancy.

Weather
Temperature, rainfall, humidity, and extreme weather events.

Income levels & vehicle ownership
Socioeconomic data influencing EV adoption and behaviour.

Electricity tariffs
Tariff structures, time-of-use rates, and pricing plans.

DATA SOURCES

JUPEM, TNB, MCMC, DOSM, OSM

JUPEM · TENAGA NASIONAL · MCMC · DOSM MALAYSIA · OpenStreetMap

QUALITY DATA. SMARTER DECISIONS. BETTER INFRASTRUCTURE.

FOUNDATIONAL KNOWLEDGE IN GEOAI

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GEOAI INTELLIGENCE

Intelligence GeoAI should generate

GEOAI TASKS



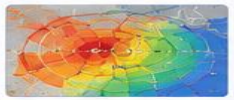
1. Demand prediction
Forecast future EV charging demand by time and location.



2. Spatial clustering (charging demand hotspots)
Identify high-demand clusters and emerging hotspot areas.



3. Suitability analysis (best candidate sites)
Evaluate locations based on multiple technical & contextual factors.



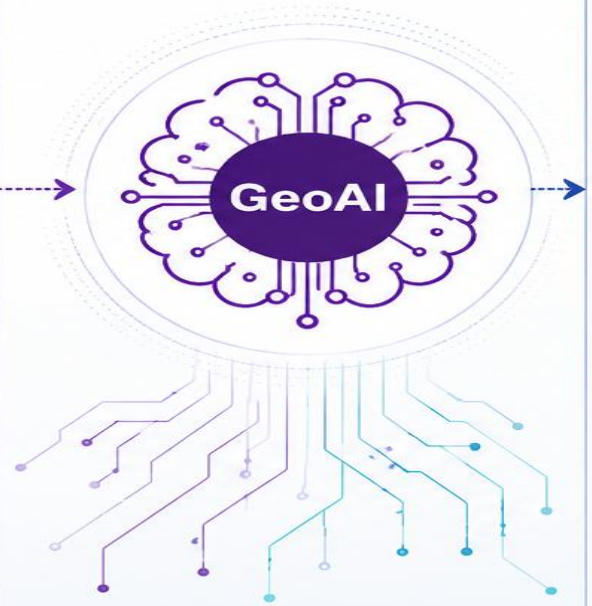
4. Accessibility modelling (travel distance & coverage)
Assess accessibility using travel time, distance and service coverage.



5. Grid capacity prediction
Predict load on the grid and identify potential capacity constraints.



6. Grid capacity prediction & optimisation of new charger placement
Optimise the number, size and location of new charging stations.



GEOAI OUTPUTS



1. Future charging demand map
Visualise forecasted demand across space and time.



2. Priority investment zones
Highlight areas with the highest need and impact for investment.



3. Recommended charging sites
Rank and propose the best locations for new chargers.



4. Expected utilisation
Estimate utilisation rates by station and time.



5. Grid impact assessment
Evaluate impact on grid capacity and identify upgrades needed.

DATA-DRIVEN DECISIONS
Leverage data & AI for smarter planning.

OPTIMISED INVESTMENT
Invest in the right places for maximum impact.

BETTER ACCESS
Improve coverage & reduce range anxiety.

EFFICIENT UTILISATION
Maximise usage and operational efficiency.

SUSTAINABLE FUTURE
Build reliable, resilient & sustainable EV infrastructure.

GeoAI transforms data into actionable intelligence for a better connected and sustainable future.

FOUNDATIONAL KNOWLEDGE IN GEOAI

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GEOAI SOLUTION DESIGN

Translate Intelligence into a Usable Solution

GeoChargeAI

An intelligent planning platform for data-driven EV charging infrastructure

SIMULATE & DECIDE:
"What happens if we build 50 new chargers?"



1. DATA INPUTS

- EV ownership & registration data
- Traffic & mobility patterns
- Land use & points of interest
- Electricity grid capacity
- Topography & elevation
- Existing charging stations

2. GEOAI SOLUTION PROCESS

DATA INTEGRATION	GEOAI ANALYTICS & MODELLING	OPTIMAL LOCATION IDENTIFICATION	SCENARIO EVALUATION	PRIORITISATION & PLANNING
Integrate multi-source geospatial and energy data	Analyse spatial patterns and predict EV charging demand	Use GeoAI to identify high-potential sites for charging stations	Evaluate scenarios for accessibility, grid readiness & impact	Rank sites and create an actionable deployment plan

GeoAI CAPABILITIES

Spatial Analysis | Demand Forecasting | Suitability Modelling | Network Analysis | Impact Assessment

3. OUTPUTS

- **Optimal EV charging station locations** — High suitability sites mapped
- **Demand & utilisation forecast** — Short, medium & long term projections
- **Accessibility improvement** — Better coverage for all communities
- **Actionable planning recommendations** — Data-driven decisions for faster implementation

KEY FEATURES

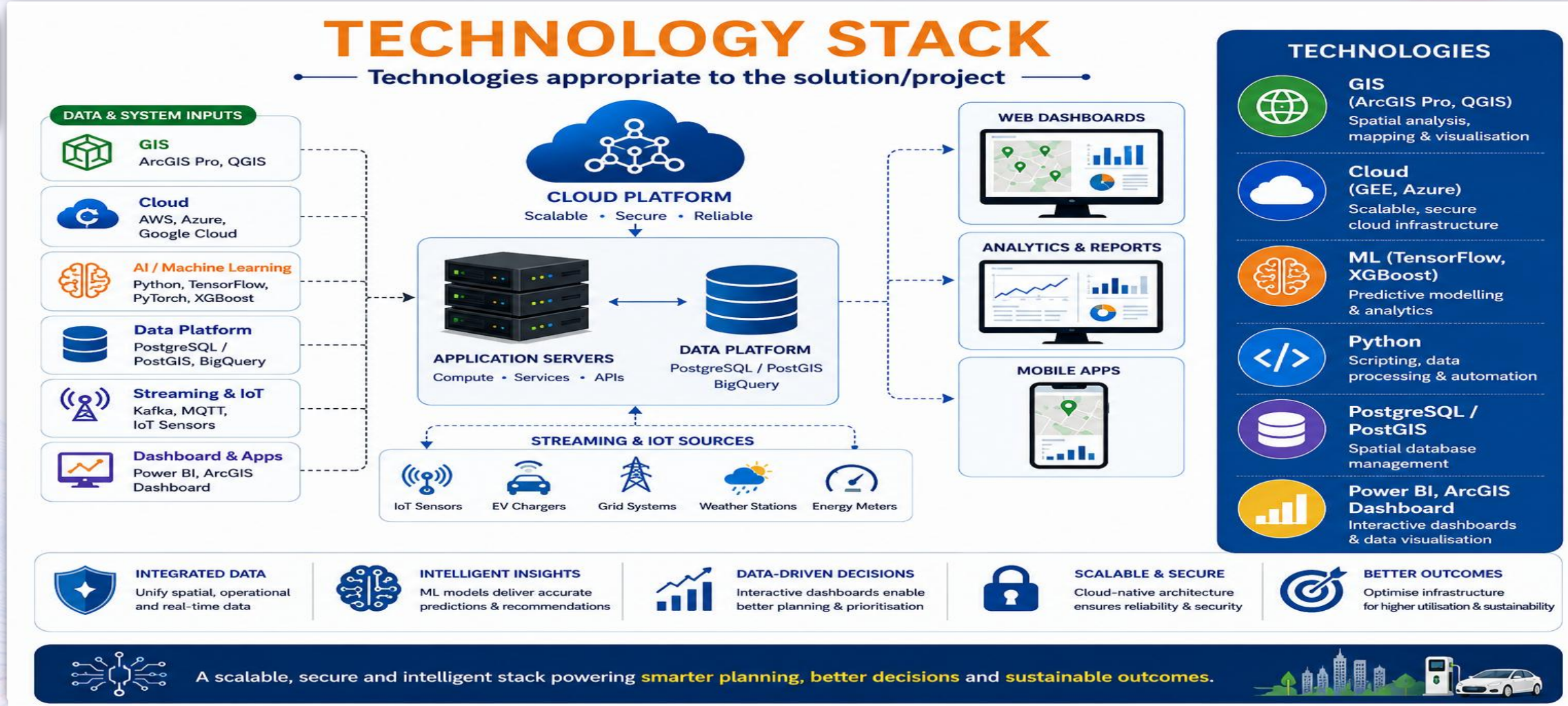
- **Interactive GIS Dashboard** — Visualise data, demand patterns and optimal sites in real time.
- **Demand Prediction** — AI models predict future EV charging demand with high accuracy.
- **Site Recommendation Engine** — Identifies the best locations based on suitability, impact and constraints.
- **Grid Capacity Monitor** — Monitor grid readiness and capacity to support new chargers.

- **DATA-DRIVEN DECISIONS** — Leverage data & AI for smarter planning.
- **OPTIMISED INVESTMENT** — Invest in the right places for maximum impact.
- **BETTER ACCESS** — Improve coverage & reduce range anxiety.
- **EFFICIENT UTILISATION** — Maximise usage and operational efficiency.
- **SUSTAINABLE FUTURE** — Build reliable, resilient & sustainable EV infrastructure.

GeoAI turns complex data into actionable intelligence for a smarter, more connected and sustainable future.

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FOUNDATIONAL KNOWLEDGE IN GEOAI

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ETHICS, PRIVACY & SUSTAINABILITY

Responsible, Ethical & Sustainable GeoAI

FAIRNESS
Ensure unbiased decisions for all users.

SOCIAL
Serve all users equitably and inclusively.

ENVIRONMENTAL
Support low-carbon future & resilience.

OPERATIONAL
Maintain reliability, transparency & governance.



SUSTAINABLE DEVELOPMENT GOALS (SDGs)

- 11 SUSTAINABLE CITIES AND COMMUNITIES — Sustainable Cities & Communities
- 13 CLIMATE ACTION — Climate Action
- 9 INDUSTRY, INNOVATION AND INFRASTRUCTURE — Industry Innovation & Infrastructure
- 17 PARTNERSHIPS FOR THE GOALS — Partnerships for Goals

X% (EXAMPLE) OF IMPACT

- **30%** Improvement in site accuracy
- **20%** Higher charger utilisation
- **25%** Reduction in wait time
- **15%** Lower carbon emissions
- CO_2 Positive impact on air quality & public health

SUPPORTING STRATEGIES & PARTNERSHIPS

- 11 COLLABORATIVE GOVERNANCE
- 13 13 CLIMATE RESILIENCE
- 9 INNOVATION FOR IMPACT
- 17 17 STRATEGIC PARTNERSHIPS for greater impact.



FOUNDATIONAL KNOWLEDGE IN GEOAI

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VALUE & IMPACT

Value the solution creates & how it will be measured

GeoChargeAI Impact

Delivering Measurable Value Across All Dimensions

ECONOMIC
Better investment decisions, lower costs

SOCIAL
Reduced range anxiety, improved accessibility

ENVIRONMENTAL
Higher EV adoption, reduced emissions

OPERATIONAL
Optimised network, improved grid planning

KPIs (Examples)

- **30%** less travel distance to a charger
- **20%** higher charger utilisation
- **25%** shorter wait times
- **15%** more EV adoption in target areas
- Annual CO_2 emissions avoided

Smarter Decisions. Better Outcomes. Sustainable Future.
GeoChargeAI empowers stakeholders to build an efficient, equitable, and low-carbon EV charging ecosystem for a better tomorrow.







GeoChargeAI creates measurable value across economic, social, environmental and operational dimensions for a smarter, more sustainable EV future.

FOUNDATIONAL KNOWLEDGE IN GEOAI

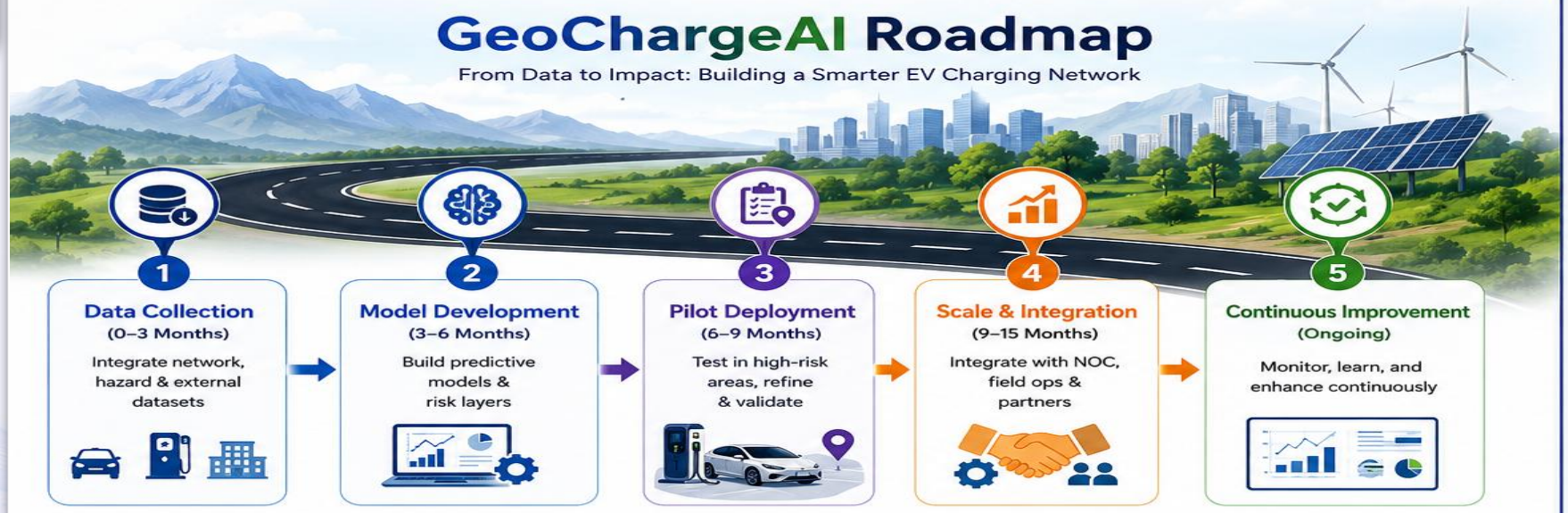
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IMPLEMENTATION ROADMAP

From pilot to full scale deployment

GeoChargeAI Roadmap

From Data to Impact: Building a Smarter EV Charging Network



1 Data Collection (0–3 Months)
Integrate network, hazard & external datasets

2 Model Development (3–6 Months)
Build predictive models & risk layers

3 Pilot Deployment (6–9 Months)
Test in high-risk areas, refine & validate

4 Scale & Integration (9–15 Months)
Integrate with NOC, field ops & partners

5 Continuous Improvement (Ongoing)
Monitor, learn, and enhance continuously

DATA-DRIVEN DECISIONS
Smarter site selection and investment planning

EFFICIENT DEPLOYMENT
Right chargers in the right places at the right time

BETTER ACCESSIBILITY
Improved coverage and user experience

LOWER EMISSIONS
Accelerating EV adoption for a cleaner future

CONTINUOUS VALUE
Adaptive models and ongoing optimisation for long-term impact

PHASES

- **Phase 1:** Data Collection
- **Phase 2:** Model Development
- **Phase 3:** Pilot Deployment
- **Phase 4:** Scale & Integration
- **Phase 5:** Continuous Improvement (Ongoing)



A clear path from pilot to full scale deployment, delivering a smarter, more resilient and future-ready EV charging network.

FOUNDATIONAL KNOWLEDGE IN GEOAI

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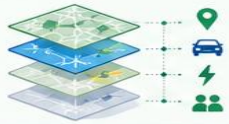
GeoChargeAI: From Data to Impact

Smarter Planning. Greater Access. A Low-Carbon Future.

1 THE PROBLEM
Uneven charging distribution limits EV adoption



2 THE DATA
Integrate geospatial, traffic, energy, demographic & EV usage data



3 THE GEOAI SOLUTION
GeoChargeAI predicts demand & identifies optimal charging sites



4 THE IMPACT
Accessible charging, efficient investment, lower emissions



5 LOW-CARBON FUTURE
Smarter cities, cleaner energy, better lives for all



1 THE PROBLEM
Uneven distribution of EV charging stations is limiting EV adoption & creating accessibility issues.



2 THE DATA
Integrate geospatial, traffic, energy, demographic & EV usage data.



3 THE GEOAI SOLUTION
GeoChargeAI predicts demand, identifies optimal charging locations & assesses grid impacts with GeoAI.



4 THE IMPACT
More accessible charging, efficient investment, reduced emissions & better urban energy planning.



5 LOW-CARBON FUTURE
Smarter cities, cleaner energy, better lives for all



MISSION 1: GEOAI BOOTCAMP 1
Building skills to solve real-world EV & sustainability challenges.

OUR GOAL
Enable data-driven decisions for an efficient, inclusive & low-carbon future.

LOW-CARBON FUTURE
- Smarter cities
- Cleaner energy
- Better lives for all



DIFFERENCE BETWEEN THE GEOAI INNOVATION CANVAS & PROBLEM-DATA-GEOSPATIAL-SOLUTION (PDGS) CANVAS

The GeoAI Innovation Canvas (GIC) develops the innovator by shaping the participants' learning journey, mindset, and entrepreneurial readiness, while the Problem-Data-Geospatial-Solution (PDGS0 Canvas, develops the innovation by systematically transforming a real-world problem into a responsible implementable, and impactful GeoAI Solution.

**" The GIC Canvas is about building the GeoAI Innovator;
the PDGS Canvas is about building the GeoAI Innovation"**

CONCLUSION

CERTIFIED ASSIGNMENT 1

CREATE YOUR OWN *PROBLEM – DATA – GEOAI SOLUTION CANVAS* USING THE 11-STEPS/DIMENSIONS APPROACH BASED ON A PROBLEM STATEMENT OF YOUR CHOICE

DO THIS ASSIGNMENT ON ONE SLIDE POWERPOINT.

UPLOAD IN THE COLAB HUB LATEST BY: 14 JULY 2026

CONCLUSION : INSPIRATIONAL FOOD FOR THOUGHT

FD 1001: FOUNDATIONAL KNOWLEDGE IN GEOAI

